

Employee Attrition Prediction – Executive Summary

Project Overview

This project analyzed employee data to identify patterns that influence employee attrition and to help the Human Resources (HR) department take proactive measures to retain valuable employees. Historical employee information, including department, job role, overtime, business travel, work-life balance, monthly income, and years of experience, was analyzed using machine learning techniques to predict whether an employee is likely to leave the company.

Key Findings

The analysis revealed that employees working as **Laboratory Technicians** and **Sales Representatives**, employees who frequently work **overtime**, and those who **travel frequently for business** have a higher likelihood of leaving the organization. The **Sales Department** experienced the highest employee attrition rate among all departments. Career-related factors such as **job level**, **total working years**, and **time since the last promotion** also had a significant influence on employee retention. Although salary contributes to employee satisfaction, the results indicate that work environment and career growth opportunities have a stronger impact on attrition than income alone.

Recommendations

To reduce employee attrition, HR should focus on employees in high-risk departments and job roles by conducting regular retention discussions, reducing excessive overtime, improving work-life balance, and providing clear career development opportunities. Regular employee engagement programs and recognition initiatives can further improve employee satisfaction and long-term retention.

Conclusion

Among the models evaluated, **Logistic Regression** provided the best overall performance for identifying employees who are at risk of leaving the company. This predictive model can help HR teams identify potential attrition early and support informed workforce planning. However, the model should be used as a decision-support tool rather than the sole basis for HR decisions, as personal circumstances and future organizational changes are not captured in the available data.